Table 5.2.2.2.4-1: Codebook for 1-layer and 2-layer CSI reporting using antenna ports 3000 to 2999+$P_{\text{CSI-RS}}$

Layers	
υ=1	$W^{(1)}_{i_{1,1},p_1^{(1)},p_1^{(2)},i_{2,1,1}} = W^{1}_{i_{1,1},p_1^{(1)},p_1^{(2)},i_{2,1,1}}$
υ=2	$W^{(2)}_{i_{1,1},p_1^{(1)},p_1^{(2)},i_{2,1,1},p_2^{(1)},p_2^{(2)},i_{2,1,2}} = \frac{1}{\sqrt{2}}\left[W^{1}_{i_{1,1},p_1^{(1)},p_1^{(2)},i_{2,1,1}} \quad W^{2}_{i_{1,1},p_2^{(1)},p_2^{(2)},i_{2,1,2}}\right]$
where $W^{l}_{i_{1,1},p_l^{(1)},p_l^{(2)},i_{2,1,l}} = \frac{1}{\sqrt{\sum_{i=0}^{2L-1}\left(p_{l,i}^{(1)}p_{l,i}^{(2)}\right)^2}}\begin{bmatrix}\sum_{i=0}^{L-1} v_{i_{1,1}d+i}\, p_{l,i}^{(1)} p_{l,i}^{(2)} \varphi_{l,i} \\ \sum_{i=0}^{L-1} v_{i_{1,1}d+i}\, p_{l,i+L}^{(1)} p_{l,i+L}^{(2)} \varphi_{l,i+L}\end{bmatrix}$, $l=1,2$, and the mappings from i_1 to $i_{1,1}$, $p_1^{(1)}$, and $p_2^{(1)}$ and from i_2 to $i_{2,1,1}$, $i_{2,1,2}$, $p_1^{(2)}$, and $p_2^{(2)}$ are as described above, including the ranges of the constituent indices of i_1 and i_2.	

5.2.2.2.5 Enhanced Type II Codebook

For 4 antenna ports {3000, 3001, …, 3003}, 8 antenna ports {3000, 3001, …, 3007}, 12 antenna ports {3000, 3001, …, 3011}, 16 antenna ports {3000, 3001, …, 3015}, 24 antenna ports {3000, 3001, …, 3023}, and 32 antenna ports {3000, 3001, …, 3031}, and UE configured with higher layer parameter *codebookType* set to ' typeII-r16'

- The values of N_1 and N_2 are configured with the higher layer parameter *n1-n2-codebookSubsetRestriction-r16*. The supported configurations of (N_1, N_2) for a given number of CSI-RS ports and the corresponding values of (O_1, O_2) are given in Table 5.2.2.2.1-2. The number of CSI-RS ports, $P_{\text{CSI-RS}}$, is $2N_1N_2$.
- The values of L, β and p_υ are determined by the higher layer parameter *paramCombination-r16*, where the mapping is given in Table 5.2.2.2.5-1.
  - The UE is not expected to be configured with *paramCombination-r16* equal to
    - 3, 4, 5, 6, 7, or 8 when $P_{\text{CSI-RS}} = 4$,
    - 7 or 8 when $P_{\text{CSI-RS}} < 32$
    - 7 or 8 when higher layer parameter *typeII-RI-Restriction-r16* is configured with $r_i = 1$ for any $i > 1$.
    - 7 or 8 when $R = 2$.
- The parameter R is configured with the higher-layer parameter *numberOfPMI-SubbandsPerCQI-Subband*. This parameter controls the total number of precoding matrices N_3 indicated by the PMI as a function of the number of configured subbands in *csi-ReportingBand*, the subband size configured by the higher-level parameter *subbandSize* and of the total number of PRBs in the bandwidth part according to Table 5.2.1.4-2, as follows:
  - When $R = 1$:
    - One precoding matrix is indicated by the PMI for each subband in *csi-ReportingBand*.
  - When $R = 2$:
    - For each subband in *csi-ReportingBand* that is not the first or last subband of a BWP, two precoding matrices are indicated by the PMI: the first precoding matrix corresponds to the first $N_{PRB}^{SB}/2$ PRBs of the subband and the second precoding matrix corresponds to the last $N_{PRB}^{SB}/2$ PRBs of the subband.
    - For each subband in *csi-ReportingBand* that is the first or last subband of a BWP
      - If $\left(N_{BWP,i}^{start} \bmod N_{PRB}^{SB}\right) \geq \frac{N_{PRB}^{SB}}{2}$, one precoding matrix is indicated by the PMI corresponding to the first subband. If $\left(N_{BWP,i}^{start} \bmod N_{PRB}^{SB}\right) < \frac{N_{PRB}^{SB}}{2}$, two precoding matrices are indicated by the PMI

corresponding to the first subband: the first precoding matrix corresponds to the first $\frac{N_{PRB}^{SB}}{2} - \left(N_{BWP,i}^{start} \bmod N_{PRB}^{SB}\right)$ PRBs of the first subband and the second precoding matrix corresponds to the last $\frac{N_{PRB}^{SB}}{2}$ PRBs of the first subband.

- If $1 + \left(N_{BWP,i}^{start} + N_{BWP,i}^{size} - 1\right) \bmod N_{PRB}^{SB} \leq \frac{N_{PRB}^{SB}}{2}$, one precoding matrix is indicated by the PMI corresponding to the last subband. If $1 + \left(N_{BWP,i}^{start} + N_{BWP,i}^{size} - 1\right) \bmod N_{PRB}^{SB} > \frac{N_{PRB}^{SB}}{2}$, two precoding matrices are indicated by the PMI corresponding to the last subband: the first precoding matrix corresponds to the first $\frac{N_{PRB}^{SB}}{2}$ PRBs of the last subband and the second precoding matrix corresponds to the last $1 + \left(N_{BWP,i}^{start} + N_{BWP,i}^{size} - 1\right) \bmod N_{PRB}^{SB} - \frac{N_{PRB}^{SB}}{2}$ PRBs of the last subband.

Table 5.2.2.2.5-1: Codebook parameter configurations for L, β and p_v

paramCombination-r16	L	p_v		β
		$v \in \{1,2\}$	$v \in \{3,4\}$	
1	2	¼	1/8	¼
2	2	¼	1/8	½
3	4	¼	1/8	¼
4	4	¼	1/8	½
5	4	¼	¼	¾
6	4	½	¼	½
7	6	¼	-	½
8	6	¼	-	¾

- The UE shall report the RI value v according to the configured higher layer parameter *typeII-RI-Restriction-r16*. The UE shall not report $v > 4$.

The PMI value corresponds to the codebook indices of i_1 and i_2 where

$$i_1 = \begin{cases} [i_{1,1} \quad i_{1,2} \quad i_{1,5} \quad i_{1,6,1} \quad i_{1,7,1} \quad i_{1,8,1}] & v = 1 \\ [i_{1,1} \quad i_{1,2} \quad i_{1,5} \quad i_{1,6,1} \quad i_{1,7,1} \quad i_{1,8,1} \quad i_{1,6,2} \quad i_{1,7,2} \quad i_{1,8,2}] & v = 2 \\ [i_{1,1} \quad i_{1,2} \quad i_{1,5} \quad i_{1,6,1} \quad i_{1,7,1} \quad i_{1,8,1} \quad i_{1,6,2} \quad i_{1,7,2} \quad i_{1,8,2} \quad i_{1,6,3} \quad i_{1,7,3} \quad i_{1,8,3}] & v = 3 \\ [i_{1,1} \quad i_{1,2} \quad i_{1,5} \quad i_{1,6,1} \quad i_{1,7,1} \quad i_{1,8,1} \quad i_{1,6,2} \quad i_{1,7,2} \quad i_{1,8,2} \quad i_{1,6,3} \quad i_{1,7,3} \quad i_{1,8,3} \quad i_{1,6,4} \quad i_{1,7,4} \quad i_{1,8,4}] & v = 4 \end{cases}$$

$$i_2 = \begin{cases} [i_{2,3,1} \quad i_{2,4,1} \quad i_{2,5,1}] & v = 1 \\ [i_{2,3,1} \quad i_{2,4,1} \quad i_{2,5,1} \quad i_{2,3,2} \quad i_{2,4,2} \quad i_{2,5,2}] & v = 2 \\ [i_{2,3,1} \quad i_{2,4,1} \quad i_{2,5,1} \quad i_{2,3,2} \quad i_{2,4,2} \quad i_{2,5,2} \quad i_{2,3,3} \quad i_{2,4,3} \quad i_{2,5,3}] & v = 3 \\ [i_{2,3,1} \quad i_{2,4,1} \quad i_{2,5,1} \quad i_{2,3,2} \quad i_{2,4,2} \quad i_{2,5,2} \quad i_{2,3,3} \quad i_{2,4,3} \quad i_{2,5,3} \quad i_{2,3,4} \quad i_{2,4,4} \quad i_{2,5,4}] & v = 4 \end{cases}$$

The precoding matrices indicated by the PMI are determined from $L + M_v$ vectors.

L vectors, $v_{m_1^{(i)}, m_2^{(i)}}$, $i = 0,1, \ldots, L-1$, are identified by the indices q_1, q_2, n_1, n_2, indicated by $i_{1,1}$, $i_{1,2}$, obtained as in 5.2.2.2.3, where the values of $C(x, y)$ are given in Table 5.2.2.2.5-4.

$M_v = \left\lceil p_v \frac{N_3}{R} \right\rceil$ vectors, $\left[y_{0,l}^{(f)}, y_{1,l}^{(f)}, \ldots, y_{N_3-1,l}^{(f)}\right]^T$, $f = 0,1, \ldots, M_v - 1$, are identified by $M_{initial}$ (for $N_3 > 19$) and $n_{3,l}$ $(l = 1, \ldots, v)$ where

$$M_{initial} \in \{-2M_v + 1, -2M_v + 2, \ldots, 0\}$$

$$n_{3,l} = \left[n_{3,l}^{(0)}, \ldots, n_{3,l}^{(M_v-1)}\right]$$

$$n_{3,l}^{(f)} \in \{0,1, \ldots, N_3 - 1\}$$

which are indicated by means of the indices $i_{1,5}$ (for $N_3 > 19$) and $i_{1,6,l}$ (for $M_\upsilon > 1$ and $l = 1, \ldots, \upsilon$), where

$$i_{1,5} \in \{0,1, \ldots, 2M_\upsilon - 1\}$$

$$i_{1,6,l} \in \begin{cases} \left\{0,1, \ldots, \binom{N_3 - 1}{M_\upsilon - 1} - 1\right\} & N_3 \leq 19 \\ \left\{0,1, \ldots, \binom{2M_\upsilon - 1}{M_\upsilon - 1} - 1\right\} & N_3 > 19 \end{cases}$$

The amplitude coefficient indicators $i_{2,3,l}$ and $i_{2,4,l}$ are

$$i_{2,3,l} = \left[k_{l,0}^{(1)} \quad k_{l,1}^{(1)}\right]$$

$$i_{2,4,l} = \left[k_{l,0}^{(2)} \ldots k_{l,M_\upsilon - 1}^{(2)}\right]$$

$$k_{l,f}^{(2)} = \left[k_{l,0,f}^{(2)} \ldots k_{l,2L-1,f}^{(2)}\right]$$

$$k_{l,p}^{(1)} \in \{1, \ldots, 15\}$$

$$k_{l,i,f}^{(2)} \in \{0, \ldots, 7\}$$

for $l = 1, \ldots, \upsilon$.

The phase coefficient indicator $i_{2,5,l}$ is

$$i_{2,5,l} = \left[c_{l,0} \ldots c_{l,M_\upsilon - 1}\right]$$

$$c_{l,f} = \left[c_{l,0,f} \ldots c_{l,2L-1,f}\right]$$

$$c_{l,i,f} \in \{0, \ldots, 15\}$$

for $l = 1, \ldots, \upsilon$.

Let $K_0 = \lceil \beta 2LM_1 \rceil$. The bitmap whose nonzero bits identify which coefficients in $i_{2,4,l}$ and $i_{2,5,l}$ are reported, is indicated by $i_{1,7,l}$

$$i_{1,7,l} = \left[k_{l,0}^{(3)} \ldots k_{l,M_\upsilon - 1}^{(3)}\right]$$

$$k_{l,f}^{(3)} = \left[k_{l,0,f}^{(3)} \ldots k_{l,2L-1,f}^{(3)}\right]$$

$$k_{l,i,f}^{(3)} \in \{0,1\}$$

for $l = 1, \ldots, \upsilon$, such that $K_l^{NZ} = \sum_{i=0}^{2L-1} \sum_{f=0}^{M_\upsilon - 1} k_{l,i,f}^{(3)} \leq K_0$ is the number of nonzero coefficients for layer $l = 1, \ldots, \upsilon$ and $K^{NZ} = \sum_{l=1}^{\upsilon} K_l^{NZ} \leq 2K_0$ is the total number of nonzero coefficients.

The indices of $i_{2,4,l}$, $i_{2,5,l}$ and $i_{1,7,l}$ are associated to the M_υ codebook indices in $n_{3,l}$.

The mapping from $k_{l,p}^{(1)}$ to the amplitude coefficient $p_{l,p}^{(1)}$ is given in Table 5.2.2.2.5-2 and the mapping from $k_{l,i,f}^{(2)}$ to the amplitude coefficient $p_{l,i,f}^{(2)}$ is given in Table 5.2.2.2.5-3. The amplitude coefficients are represented by

$$p_l^{(1)} = \left[p_{l,0}^{(1)} \quad p_{l,1}^{(1)}\right]$$

$$p_l^{(2)} = \left[p_{l,0}^{(2)} \ldots p_{l,M_\upsilon - 1}^{(2)}\right]$$

$$p_{l,f}^{(2)} = \left[p_{l,0,f}^{(2)} \ldots p_{l,2L-1,f}^{(2)}\right]$$

for $l = 1, \ldots, \upsilon$.

Let $f_l^* \in \{0,1,\ldots,M_v-1\}$ be the index of $i_{2,4,l}$ and $i_l^* \in \{0,1,\ldots,2L-1\}$ be the index of $k_{l,f_l^*}^{(2)}$ which identify the strongest coefficient of layer l, i.e., the element $k_{l,i_l^*,f_l^*}^{(2)}$ of $i_{2,4,l}$, for $l = 1,\ldots,v$. The codebook indices of $n_{3,l}$ are remapped with respect to $n_{3,l}^{(f_l^*)}$ as $n_{3,l}^{(f)} = \left(n_{3,l}^{(f)} - n_{3,l}^{(f_l^*)}\right) \bmod N_3$, such that $n_{3,l}^{(f_l^*)} = 0$, after remapping. The index f is remapped with respect to f_l^* as $f = (f - f_l^*) \bmod M_v$, such that the index of the strongest coefficient is $f_l^* = 0$ ($l = 1,\ldots,v$), after remapping. The indices of $i_{2,4,l}$, $i_{2,5,l}$ and $i_{1,7,l}$ indicate amplitude coefficients, phase coefficients and bitmap after remapping.

The strongest coefficient of layer l is identified by $i_{1,8,l} \in \{0,1,\ldots,2L-1\}$, which is obtained as follows

$$i_{1,8,l} = \begin{cases} \sum_{i=0}^{i_1^*} k_{1,i,0}^{(3)} - 1 & v = 1 \\ i_l^* & 1 < v \le 4 \end{cases}$$

for $l = 1,\ldots,v$.

Table 5.2.2.2.5-2: Mapping of elements of $i_{2,3,l}$: $k_{l,p}^{(1)}$ to $p_{l,p}^{(1)}$

$k_{l,p}^{(1)}$	$p_{l,p}^{(1)}$	$k_{l,p}^{(1)}$	$p_{l,p}^{(1)}$	$k_{l,p}^{(1)}$	$p_{l,p}^{(1)}$	$k_{l,p}^{(1)}$	$p_{l,p}^{(1)}$
0	Reserved	4	$\left(\frac{1}{2048}\right)^{1/4}$	8	$\left(\frac{1}{128}\right)^{1/4}$	12	$\left(\frac{1}{8}\right)^{1/4}$
1	$\frac{1}{\sqrt{128}}$	5	$\frac{1}{2\sqrt{8}}$	9	$\frac{1}{\sqrt{8}}$	13	$\frac{1}{\sqrt{2}}$
2	$\left(\frac{1}{8192}\right)^{1/4}$	6	$\left(\frac{1}{512}\right)^{1/4}$	10	$\left(\frac{1}{32}\right)^{1/4}$	14	$\left(\frac{1}{2}\right)^{1/4}$
3	$\frac{1}{8}$	7	$\frac{1}{4}$	11	$\frac{1}{2}$	15	1

The amplitude and phase coefficient indicators are reported as follows:

- $k_{l,\left\lfloor \frac{i_l^*}{L} \right\rfloor}^{(1)} = 15$, $k_{l,i_l^*,0}^{(2)} = 7$, $k_{l,i_l^*,0}^{(3)} = 1$ and $c_{l,i_l^*,0} = 0$ $(l = 1,\ldots,v)$. The indicators $k_{l,\left\lfloor \frac{i_l^*}{L} \right\rfloor}^{(1)}$, $k_{l,i_l^*,0}^{(2)}$ and $c_{l,i_l^*,0}$ are not reported for $l = 1,\ldots,v$.
- The indicator $k_{l,\left(\left\lfloor \frac{i_l^*}{L} \right\rfloor + 1\right) mod\ 2}^{(1)}$ is reported for $l = 1,\ldots,v$.
- The $K^{NZ} - v$ indicators $k_{l,i,f}^{(2)}$ for which $k_{l,i,f}^{(3)} = 1$, $i \ne i_l^*$, $f \ne 0$ are reported.
- The $K^{NZ} - v$ indicators $c_{l,i,f}$ for which $k_{l,i,f}^{(3)} = 1$, $i \ne i_l^*$, $f \ne 0$ are reported.
- The remaining $2L \cdot M_v \cdot v - K^{NZ}$ indicators $k_{l,i,f}^{(2)}$ are not reported.
- The remaining $2L \cdot M_v \cdot v - K^{NZ}$ indicators $c_{l,i,f}$ are not reported.

Table 5.2.2.2.5-3: Mapping of elements of $i_{2,4,l}$: $k_{l,i,f}^{(2)}$ to $p_{l,i,f}^{(2)}$

$k_{l,i,f}^{(2)}$	$p_{l,i,f}^{(2)}$

0	$\frac{1}{8\sqrt{2}}$
1	$\frac{1}{8}$
2	$\frac{1}{4\sqrt{2}}$
3	$\frac{1}{4}$
4	$\frac{1}{2\sqrt{2}}$
5	$\frac{1}{2}$
6	$\frac{1}{\sqrt{2}}$
7	1

The elements of n_1 and n_2 are found from $i_{1,2}$ using the algorithm described in 5.2.2.2.3, where the values of $C(x, y)$ are given in Table 5.2.2.2.5-4.

For $N_3 > 19$, $M_{initial}$ is identified by $i_{1,5}$.

For all values of N_3, $n_{3,l}^{(0)} = 0$ for $l = 1, \ldots, \upsilon$. If $M_\upsilon > 1$, the nonzero elements of $n_{3,l}$, identified by $n_{3,l}^{(1)}, \ldots, n_{3,l}^{(M_\upsilon - 1)}$, are found from $i_{1,6,l}$ $(l = 1, \ldots, \upsilon)$, for $N_3 \leq 19$, and from $i_{1,6,l}$ $(l = 1, \ldots, \upsilon)$ and $M_{initial}$, for $N_3 > 19$, using $C(x, y)$ as defined in Table 5.2.2.2.5-4 and the algorithm:

$s_0 = 0$

for $f = 1, \ldots, M_\upsilon - 1$

Find the largest $x^* \in \{M_\upsilon - 1 - f, \ldots, N_3 - 1 - f\}$ in Table 5.2.2.2.5-4 such that

$i_{1,6,l} - s_{f-1} \geq C(x^*, M_\upsilon - f)$

$e_f = C(x^*, M_\upsilon - f)$

$s_f = s_{f-1} + e_f$

if $N_3 \leq 19$

$n_{3,l}^{(f)} = N_3 - 1 - x^*$

else

$n_l^{(f)} = 2M_\upsilon - 1 - x^*$

if $n_l^{(f)} \leq M_{initial} + 2M_\upsilon - 1$

$n_{3,l}^{(f)} = n_l^{(f)}$

else

$n_{3,l}^{(f)} = n_l^{(f)} + (N_3 - 2M_\upsilon)$

end if

end if

Table 5.2.2.2.5-4: Combinatorial coefficients $C(x, y)$

x \ y	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0
2	2	1	0	0	0	0	0	0	0
3	3	3	1	0	0	0	0	0	0
4	4	6	4	1	0	0	0	0	0
5	5	10	10	5	1	0	0	0	0
6	6	15	20	15	6	1	0	0	0
7	7	21	35	35	21	7	1	0	0
8	8	28	56	70	56	28	8	1	0
9	9	36	84	126	126	84	36	9	1
10	10	45	120	210	252	210	120	45	10
11	11	55	165	330	462	462	330	165	55
12	12	66	220	495	792	924	792	495	220
13	13	78	286	715	1287	1716	1716	1287	715
14	14	91	364	1001	2002	3003	3432	3003	2002
15	15	105	455	1365	3003	5005	6435	6435	5005
16	16	120	560	1820	4368	8008	11440	12870	11440
17	17	136	680	2380	6188	12376	19448	24310	24310
18	18	153	816	3060	8568	18564	31824	43758	48620

When $n_{3,l}$ and $M_{initial}$ are known, $i_{1,5}$ and $i_{1,6,l}$ $(l = 1, \ldots, \upsilon)$ are found as follows:

- If $N_3 \leq 19$, $i_{1,5} = 0$ and is not reported. If $M_\nu = 1$, $i_{1,6,l} = 0$, for $l = 1, \ldots, \nu$, and is not reported. If $M_\nu > 1$, $i_{1,6,l} = \sum_{f=1}^{M_\upsilon - 1} C\left(N_3 - 1 - n_{3,l}^{(f)}, M_\upsilon - f\right)$, where $C(x, y)$ is given in Table 5.2.2.2.5-4 and where the indices $f = 1, \ldots, M_\upsilon - 1$ are assigned such that $n_{3,l}^{(f)}$ increases as f increases.

- If $N_3 > 19$, $M_{initial}$ is indicated by $i_{1,5}$, which is reported and given by

$$i_{1,5} = \begin{cases} M_{initial} & M_{initial} = 0 \\ M_{initial} + 2M_\upsilon & M_{initial} < 0 \end{cases}$$

  Only the nonzero indices $n_{3,l}^{(f)} \in IntS$, where $IntS = \{(M_{initial} + i) \bmod N_3, i = 0,1, \ldots, 2M_\upsilon - 1\}$, are reported, where the indices $f = 1, \ldots, M_\upsilon - 1$ are assigned such that $n_{3,l}^{(f)}$ increases as f increases. Let

$$n_l^{(f)} = \begin{cases} n_{3,l}^{(f)} & n_{3,l}^{(f)} \leq M_{initial} + 2M_\upsilon - 1 \\ n_{3,l}^{(f)} - (N_3 - 2M_\upsilon) & n_{3,l}^{(f)} > M_{initial} + N_3 - 1 \end{cases},$$

  then $i_{1,6,l} = \sum_{f=1}^{M_\upsilon - 1} C\left(2M_\upsilon - 1 - n_l^{(f)}, M_\upsilon - f\right)$, where $C(x, y)$ is given in Table 5.2.2.2.5-4.

The codebooks for 1-4 layers are given in Table 5.2.2.2.5-5, where $m_1^{(i)}, m_2^{(i)}$, for $i = 0,1, \ldots, L - 1$, $v_{m_1^{(i)}, m_2^{(i)}}$ are obtained as in clause 5.2.2.2.3, and the quantities $\varphi_{l,i,f}$ and $y_{t,l}$ are given by

$$\varphi_{l,i,f} = e^{j\frac{2\pi c_{l,i,f}}{16}}$$

$$y_{t,l} = \left[y_{t,l}^{(0)} \quad y_{t,l}^{(1)} \quad \ldots \quad y_{t,l}^{(M_\upsilon - 1)}\right]$$

where $t = \{0,1, \ldots, N_3 - 1\}$, is the index associated with the precoding matrix, $l = \{1, \ldots, \upsilon\}$, and with

$$y_{t,l}^{(f)} = e^{j\frac{2\pi t n_{3,l}^{(f)}}{N_3}}$$

for $f = 0,1,\ldots,M_\upsilon - 1$.

Table 5.2.2.2.5-5: Codebook for 1-layer. 2-layer, 3-layer and 4-layer CSI reporting using antenna ports 3000 to 2999+$P_{\text{CSI-RS}}$

Layers	
$\upsilon = 1$	$W^{(1)}_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},t} = W^1_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},t}$
$\upsilon = 2$	$W^{(2)}_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},n_{3,2},p_2^{(1)},p_2^{(2)},i_{2,5,2},t} = \frac{1}{\sqrt{2}}\left[W^1_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},t}\ W^2_{q_1,q_2,n_1,n_2,n_{3,2},p_2^{(1)},p_2^{(2)},i_{2,5,2},t}\right]$
$\upsilon = 3$	$W^{(3)}_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},n_{3,2},p_2^{(1)},p_2^{(2)},i_{2,5,2},n_{3,3},p_3^{(1)},p_3^{(2)},i_{2,5,3},t} = \frac{1}{\sqrt{3}}\left[W^1_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},t}\ W^2_{q_1,q_2,n_1,n_2,n_{3,2},p_2^{(1)},p_2^{(2)},i_{2,5,2},t}\ W^3_{q_1,q_2,n_1,n_2,n_{3,3},p_3^{(1)},p_3^{(2)},i_{2,5,3},t}\right]$
$\upsilon = 4$	$W^{(4)}_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},n_{3,2},p_2^{(1)},p_2^{(2)},i_{2,5,2},n_{3,3},p_3^{(1)},p_3^{(2)},i_{2,5,3},n_{3,4},p_4^{(1)},p_4^{(2)},i_{2,5,4},t} = \frac{1}{2}\left[W^1_{q_1,q_2,n_1,n_2,n_{3,1},p_1^{(1)},p_1^{(2)},i_{2,5,1},t}\ W^2_{q_1,q_2,n_1,n_2,n_{3,2},p_2^{(1)},p_2^{(2)},i_{2,5,2},t}\ W^3_{q_1,q_2,n_1,n_2,n_{3,3},p_3^{(1)},p_3^{(2)},i_{2,5,3},t}\ W^4_{q_1,q_2,n_1,n_2,n_{3,4},p_4^{(1)},p_4^{(2)},i_{2,5,4},t}\right]$
Where $W^l_{q_1,q_2,n_1,n_2,n_{3,l},p_l^{(1)},p_l^{(2)},i_{2,5,l},t} = \frac{1}{\sqrt{N_1N_2\gamma_{t,l}}}\begin{bmatrix}\sum_{i=0}^{L-1} v_{m_1^{(i)},m_2^{(i)}} p_{l,0}^{(1)} \sum_{f=0}^{M_\upsilon-1} y_{t,l}^{(f)} p_{l,i,f}^{(2)} \varphi_{l,i,f} \\ \sum_{i=0}^{L-1} v_{m_1^{(i)},m_2^{(i)}} p_{l,1}^{(1)} \sum_{f=0}^{M_\upsilon-1} y_{t,l}^{(f)} p_{l,i+L,f}^{(2)} \varphi_{l,i+L,f}\end{bmatrix}$, $l = 1,2,3,4$, $\gamma_{t,l} = \sum_{i=0}^{2L-1}\left(p_{l,\lfloor\frac{i}{L}\rfloor}^{(1)}\right)^2 \left\lvert\sum_{f=0}^{M_\upsilon-1} y_{t,l}^{(f)} p_{l,i,f}^{(2)} \varphi_{l,i,f}\right\rvert^2$ and the mappings from i_1 to q_1, q_2, n_1, n_2, $n_{3,1}$, $n_{3,2}$, $n_{3,3}$, $n_{3,4}$, and from i_2 to $i_{2,5,1}$, $i_{2,5,2}$, $i_{2,5,3}$, $i_{2,5,4}$, $p_1^{(1)}$, $p_2^{(1)}$, $p_3^{(1)}$ and $p_4^{(1)}$, $p_1^{(2)}$, $p_2^{(2)}$, $p_3^{(2)}$ and $p_4^{(2)}$ are as described above, including the ranges of the constituent indices of i_1 and i_2.	

For coefficients with $k_{l,i,f}^{(3)} = 0$, amplitude and phase are set to zero, i.e., $p_{l,i,f}^{(2)} = 0$ and $\varphi_{l,i,f} = 0$.

The bitmap parameter *typeII-RI-Restriction-r16* forms the bit sequence r_3, r_2, r_1, r_0 where r_0 is the LSB and r_3 is the MSB. When r_i is zero, $i \in \{0,1,\ldots,3\}$, PMI and RI reporting are not allowed to correspond to any precoder associated with $\upsilon = i + 1$ layers.

The bitmap parameter *n1-n2-codebookSubsetRestriction-r16* forms the bit sequence $B = B_1B_2$ and configures the vector group indices $g^{(k)}$ as in clause 5.2.2.2.3. Bits $b_2^{(k,2(N_1x_2+x_1)+1)}b_2^{(k,2(N_1x_2+x_1))}$ indicate the maximum allowed average amplitude, γ_{i+pL} $(p = 0,1)$, with $i \in \{0,1,\ldots,L-1\}$, of the coefficients associated with the vector in group $g^{(k)}$ indexed by x_1, x_2, where the maximum amplitudes are given in Table 5.2.2.2.5-6 and the average coefficient amplitude is restricted as follows

$$\sqrt{\frac{1}{\sum_{f=0}^{M_\upsilon-1} k_{l,i+pL,f}^{(3)}} \sum_{f=0}^{M_\upsilon-1} k_{l,i+pL,f}^{(3)} \left(p_{l,p}^{(1)} p_{l,i+pL,f}^{(2)}\right)^2} \le \gamma_{i+pL}$$

for $l = 1,\ldots,\upsilon$, and $p = 0,1$. A UE that does not report the parameter *amplitudeSubsetRestriction-r16* = 'supported' in its capability signaling is not expected to be configured with $b_2^{(k,2(N_1x_2+x_1)+1)}b_2^{(k,2(N_1x_2+x_1))} = 01$ or 10.

Table 5.2.2.2.5-6: Maximum allowed average coefficient amplitudes for restricted vectors

Bit $b_2^{(k,2(N_1x_2+x_1)+1)}b_2^{(k,2(N_1x_2+x_1))}$	Maximum Average Coefficient Amplitude γ_{i+pL}

00	0
01	$\sqrt{1/4}$
10	$\sqrt{1/2}$
11	1

5.2.2.2.5a Refined eType II Codebook

For 48, 64 and 128 antenna ports, obtained by aggregating K CSI-RS resources for channel measurement, and the UE configured with higher layer parameter *codebookType* set to 'eTypeII-r19'

- The values of N_1 and N_2 are configured with the higher layer parameter *n1-n2-typeII-r19*. The supported configurations of (N_1, N_2), for a given number of CSI-RS ports, are given in Table 5.2.2.2.1a-1, for which $O_1 = O_2 = 4$. The number of CSI-RS ports, P_{CSI-RS}, is $2N_1N_2$.
- The port index, $p' = 0,1, \ldots, P_{CSI-RS} - 1$, is defined in Clause 7.4.1.5.3 of [4, TS 38.211].
- The values of L, β and p_v are determined by the higher layer parameter *paramCombination-r19*, where the same mapping is used as in Table 5.2.2.2.5-1.
  - The UE is not expected to be configured with *paramCombination-r19* equal to
    - 7 or 8 when higher layer parameter t*ypeII-RI-Restriction-r19* is configured with $r_i = 1$ for any $i > 1$.
    - 7 or 8 when $R = 2$.
- The parameter R is configured with the higher layer parameter *numberOfPMI-SubbandsPerCQI-subband-r19* and defined as in Clause 5.2.2.2.5.
- The UE shall report the RI value v according to the configured higher layer parameter *typeII-RI-restriction-r19*, as described in Clause 5.2.2.2.5. The UE shall not report $v > 4$.

The codebook is defined as in Clause 5.2.2.2.5. L vectors, $v_{m_1^{(i)}, m_2^{(i)}}$, $i = 0,1, \ldots, L-1$, are identified by the indices q_1, q_2, n_1, n_2, indicated by $i_{1,1}$, $i_{1,2}$, obtained as in 5.2.2.2.3, where the values of $C(x, y)$ are given in Table 5.2.2.2.5-4 and 5.2.2.2.1a-5.

The bitmap parameter *typeII-CBSR-r19* forms the bit sequence $a_{A_c-1}, \ldots, a_1, a_0$, where a_0 is the LSB and a_{A_c-1} is the MSB and the number of bits is given by $A_c = N_1N_2/\left(X_1^{(c)}X_2^{(c)}\right)$, where the values $\left(X_1^{(c)}, X_2^{(c)}\right)$ are configured by the higher layer parameters *valueOfX1-typeII-CBSR-r19* and *valueOfX2-typeII-CBSR-r19*, respectively. The supported combinations of $\left(X_1^{(c)}, X_2^{(c)}\right)$ for a given combination of (N_1, N_2) are given in Table 5.2.2.2.5a-1. A bit value of zero for bit $a_{\frac{N_2}{X_2^{(c)}}x_1+x_2}$ indicates that PMI reporting is not allowed to correspond to any precoder associated with vector group $G_a(x_1, x_2)$, $x_1 = 0, \ldots, \frac{N_1}{X_1^{(c)}} - 1$, $x_2 = 0, \ldots, \frac{N_2}{X_2^{(c)}} - 1$, given by

$$G_a(x_1, x_2) = \left\{v_{X_1^{(c)}x_1+l, X_2^{(c)}x_2+m} : l = 0,1, \ldots, X_1^{(c)} - 1; m = 0,1, \ldots, X_2^{(c)} - 1\right\}.$$

Table 5.2.2.2.5a-1: Supported combinations of $(X_1^{(c)}, X_2^{(c)})$ for codebook subset restriction

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)	$(X_1^{(c)}, X_2^{(c)})$
48	(8,3)	(1,1), (2,1), (4,1)
	(6,4)	(1,1), (2,1), (2,2)
64	(16,2)	(1,1), (2,1), (2,2), (4,1), (4,2)
	(8,4)	(1,1), (2,1), (2,2), (4,1), (4,2)
128	(16,4)	(1,1), (2,1), (2,2), (4,1), (4,2)
	(8,8)	(1,1), (2,1), (2,2), (4,1), (4,2)